GestureForge Assignment Design Report Draft

Status: formal draft, not ready to submit. Documentation audited against the repository and recorded evidence on 27 September 2026; the recorded hardware and pilot runs are dated 23 September. The prototype has software checks, a development-validation comparison and a participant-observed two-hold NEXT smoke check. Fresh-source setup was tested on the development computer only. Assignment completion depends on the open assignment checklist gates: final report review, second-computer acceptance, tutor/team review, private P01 package retention notes, AI-use guidance check, signed declaration, and LMS upload/download verification. Formal repeated-hold, live false-action, latency and held-out evaluation remain final-project deliverables.

Names supplied by the user: **Jameson and Deston**. Full names, student IDs, university emails, group name, tutorial/registration, verified individual work and agreed percentages remain pending. GitHub and WhatsApp use was confirmed on 27 September; no tutor approval or signed declaration is recorded.

Executive Summary

GestureForge is a local Windows desktop AI prototype for controlling a small card deck with personal static hand gestures. The assignment release proves the core AI subsystem and one real product workflow: webcam input is converted to MediaPipe hand landmarks, normalised into 60 numeric features, classified by both kNN and RBF SVM for comparison, filtered through distance-based rejection, and passed to a hold-and-rearm controller that executes NEXT once per deliberate hold.

The assignment scope is deliberately narrow so it can be demonstrated and defended. NEXT is the only gesture that performs a viewer action. PREVIOUS and BOOKMARK are recognised and displayed as preview-only labels, leaving meaningful implementation work for the 7 November final project: in-app enrolment, multiple profiles, PREVIOUS and BOOKMARK execution, user-selected validated JSON decks, saved state, held-out testing for two participants, and a final demo video.

The prototype is strong enough for a near-final assignment build but not ready to submit until the remaining human and packaging gates are closed. Software verification currently passes Ruff, mypy, pytest, GUI smoke launch and synthetic evaluation. The P01 right-hand pilot collected separate TRAIN and VALIDATION sessions with 32 trials and 160 valid scheduled samples each, plus 40 valid validation-negative samples. Corrected development validation reached macro-F1 0.9686 for both kNN and SVM at rejection multiplier 2.0, with 31 of 32 validation trials correct. Live P01 testing advanced cards through the full webcam-to-viewer NEXT path and confirmed two sustained NEXT holds each fired exactly once after NEUTRAL rearming.

P02 laptop feedback recorded on 26 September reports successful ZIP use with Codex assistance and a separately collected right-hand dataset: NEXT advanced Card 1 to Card 10, a five-second NEXT hold produced one action, NEUTRAL rearm was required, and PREVIOUS/BOOKMARK remained preview-only. The participant reported reduced reliability with motion, tilt, back-of-hand orientation, lateral position and excessive distance. These are qualitative observations, not isolated root causes or a formal guide-only installation, disconnected-network, pause/reopen or quantitative P02 accuracy result.

Problem and User Need

The target user is a presenter, learner, or demonstrator who wants to move through a small deck without touching the keyboard or mouse. The problem is not general sign-language recognition. The problem is a bounded human-computer interaction task: recognise a small personal command set while avoiding accidental navigation.

The prototype uses personal examples rather than a universal gesture model. That keeps the assignment feasible because the classifier only needs to distinguish one enrolled user's static labels under documented camera, lighting, hand, and distance conditions.

Innovation and AI Contribution

Common low-risk alternatives for this workflow are keyboard shortcuts, clickers, touch screens, voice commands, or fixed rule-based hand-pose thresholds. GestureForge's innovation is the combination of personal gesture enrolment, a small explainable classifier comparison, open-set rejection, and deliberate activation control. The aim is not to replace high-accuracy sign-language systems; it is to make a small touch-free viewer that can be trained quickly for one user and made safer against accidental repeated actions.

MediaPipe supplies the pretrained hand landmarks. The project's AI contribution is the system built around those landmarks: the dataset and manifest design, 60-feature normalisation, train-only scaling, kNN and SVM comparison, class-distance rejection, validation-negative analysis, and controller integration with hold/rearm safeguards. A real webcam NEXT gesture advances a card in the application; this integration result does not establish general live safety. AI assistance and verified human contributions must be disclosed separately.

Assignment Scope

Implemented for the assignment prototype:

- Command-line collection for NEXT, PREVIOUS, BOOKMARK, NEUTRAL, and validation-negative sequences.
- Versioned manifest and numeric feature storage without pickle or joblib model files.
- Feature extraction from 21 MediaPipe landmarks into 60 normalised features.
- kNN and SVM training with train-only scaling.
- Class-conditional distance rejection.
- Reproducible evaluation from JSON config.
- A PySide6 desktop shell with fixed deck, camera preview area, prediction display, controller status, and keyboard/button fallback.
- A controller that can execute NEXT once per deliberate hold.
- Clear assignment boundary text showing PREVIOUS and BOOKMARK as preview-only.
- Local agile board, setup helper, source packaging helper, and documentation set.

Still pending before assignment submission:

- Human review of the exported report draft; regenerate after any content changes.
- Second Windows computer acceptance check from the laptop guide; same-computer archive checks do not close it.
- Tutor/team review, full student identifiers, and contribution confirmation.
- Private P01 package retention note and AI-use guidance check.
- Signed group declaration and LMS upload/download verification.

Excluded until the final project:

- In-app enrolment and practice.
- Profile create, reload, and delete.
- PREVIOUS and BOOKMARK execution.
- User-selected, validated JSON deck loading; the assignment already loads its fixed ten-card deck from JSON.
- Saved deck position and bookmarks.
- Two-participant held-out final evaluation.

Not in agreed scope: PowerPoint control, operating-system control, dynamic gestures, two-handed gestures, sign-language translation, cloud accounts, mobile apps, deep-network training, or an installer executable.

Systems Engineering Approach

The project uses a tailored lifecycle. NASA's lifecycle model separates concept, preliminary design, detailed design, integration and test, operations, and closeout through gated reviews. GestureForge adapts that structure at assignment scale:

- Concept and feasibility: assignment scope, constraints, risk register, and assessment mapping.

- Preliminary design: requirements, data contracts, feature extraction, classifier plan, controller plan, and report draft.

- Detailed design and build: collection CLI, dataset persistence, training and evaluation, UI shell, controller, viewer, and helper scripts.

- Integration and verification: unit tests, static checks, GUI smoke test, model asset smoke test, live camera check, and release packaging.

- Operation and closeout: user guide, AI-use record, contribution record, signed declaration, release package, and final-project handover notes.

Decisions D-001 to D-007 and risks R-001 to R-006 in DECISIONS_AND_RISKS.md connect scope, calibration, privacy and release choices to their evidence. Pending tutor gates are not presented as completed lifecycle reviews.

Agile tracking is maintained through docs/backlog.csv, local board generation with python scripts/board.py, and the private GitHub repository and project board recorded in docs/GITHUB_SETUP.md.

Milestones

The schedule keeps the assignment prototype and final project clearly separated.

Date range	Assignment milestone	Exit gate
23-24 Sep	Repository layout, dependency check, camera/landmark feasibility and requirements draft	Development machine can run the stack and capture usable webcam input
25-26 Sep	Collection CLI, feature pipeline, manifest, P01 pilot data and default training	Saved numeric data and first classifier outputs exist
27-28 Sep	Fixed viewer, controller and live prediction integration	Real webcam NEXT advances one card once
29-30 Sep	Classifier comparison, rejection tuning, tests and measured outputs	Preliminary results are reproducible and limitations are recorded
1 Oct	Prototype freeze, install instructions and report reconciliation	Report claims match code and evidence
2 Oct	Package review, declaration and LMS upload check	Downloaded submission package is rechecked
3 Oct	Deadline buffer	Fixes and packaging corrections only

Date range	Final project milestone	Exit gate
4-11 Oct	In-app enrolment, practice, profile lifecycle and validation	Profiles can be created and reloaded
12-18 Oct	PREVIOUS, BOOKMARK, JSON decks, saved state and recovery	Feature freeze
19-25 Oct	Bounded validation search, frozen candidates, held-out data and live tasks	Final evaluation data collected
26 Oct-1 Nov	Defect fixes, second-machine release test and stability run	Release candidate
2-6 Nov	Final report, guide, video, slides and rehearsal	Submission ready
7 Nov	Deadline buffer	No new features

Requirements and Current Status

ID	Requirement	Assignment status	Evidence or blocker
RQ-001	The system shall run locally on Windows with Python 3.11.	Software verified; assisted laptop use reported	Python 3.11.16 development environment recorded; P02 package/version details and full release acceptance remain pending
RQ-002	The system shall store project work under code.	Implemented	Repository and artifacts are under code
RQ-003	The prototype shall collect labelled examples for NEXT, PREVIOUS, BOOKMARK, and NEUTRAL.	Implemented, pilot collected	python -m gestureforge.collect; P01 right-hand TRAIN and VALIDATION each have 32 trials and 160/160 valid scheduled samples; validation-negative has 40 valid samples
RQ-004	The feature extractor shall produce 60 normalised landmark features from one selected hand.	Software verified	Automated tests reported passed
RQ-005	The system shall reject no hand, multiple hands, wrong hand, non-finite landmarks, and degenerate scale.	Software verified	Automated tests and blank-image asset smoke reported
RQ-006	The assignment shall compare kNN and SVM using the same preprocessing policy.	Development validation complete	python -m gestureforge.evaluate; corrected kNN and SVM share the same aggregate macro-F1 values across tested multipliers
RQ-007	The controller shall require neutral arming and command hold before action dispatch.	Software verified	Controller tests reported passed
RQ-008	The assignment viewer shall execute NEXT only.	Verified for basic live path	Genuine webcam gesture NEXT events advanced the viewer to positions 1, 2, and 3; participant-observed two-hold smoke check advanced once per hold
RQ-009	The viewer shall avoid duplicate command execution from one continuous hold.	Software verified	Controller and viewer tests reported passed
RQ-010	PREVIOUS and BOOKMARK shall be displayed as recognition preview only in the assignment.	Implemented	UI boundary text present
RQ-011	Dataset manifests shall keep invalid detections rather than dropping them silently.	Software verified	Dataset tests reported passed
RQ-012	Evaluation shall report macro-F1, per-class precision and recall, and confusion including UNKNOWN.	Development validation complete	Aggregate pilot metrics recorded in docs/evidence/PILOT_RESULTS.md; not held-out final test evidence
RQ-013	The final project shall support profile persistence and reload.	Future	Not assignment scope
RQ-014	The final project shall support user-selected, validated JSON decks.	Future	Assignment uses only the fixed bundled JSON deck
RQ-015	The final project shall include complete guide, video, and presentation evidence.	Future	Not assignment scope

Architecture

```
Webcam -> MediaPipe landmarks -> 60 normalised features -> kNN/SVM -> distance rejection -> activation controller -> fixed card viewer -> event log
```

The camera and inference worker run away from direct widget updates. The UI consumes only the latest available result, so a slow interface should not build an old-frame queue. Preview frames are mirrored for the user, but feature extraction uses the unmirrored landmark convention.

Subsystem Inputs and Outputs

Subsystem	Input	Output	Main safeguard
Capture	Webcam frame and MediaPipe model asset	Observation with hand count, landmarks, handedness, timestamp	No raw frame recording
Feature extraction	One valid selected-hand observation	60-feature vector	Reject invalid scale, wrong hand, no hand, multiple hands, non-finite values
Dataset	Sample records and feature matrix	JSON manifest and NPZ sidecar	Hash check and split integrity validation
Model training	Training features and labels	In-memory prototype model config	Train-only StandardScaler; no executable model persistence
Evaluation	JSON config and dataset	Metrics, confusion matrix, predictions, timing	UNKNOWN accounting and trial-level majority
Controller	Accepted predictions and current time	Zero or one command event	Neutral arm, command hold, stale-frame rejection, latch
Viewer	Command event or keyboard input	Updated fixed deck position	No wrap and duplicate event rejection
Release tooling	Tracked Git files; optional authorised P01 data	Source ZIP or separate private marker ZIP and checksum	Default excludes private data; explicit option validates and adds only the P01 manifest and numeric sidecar

Data Contracts

The code defines Observation, Prediction, and CommandEvent dataclasses in contracts.py. Dataset records are stored as reviewable JSON records with a compressed NumPy feature sidecar. The live path carries session, profile and frame IDs from observations through predictions to gesture events. This is runtime identity linkage, not a persisted history of every observation or prediction; manual navigation is logged separately with source keyboard.

The assignment does not persist trained pickle or joblib models. It saves numeric training data and JSON configuration, then rebuilds the small classifiers when needed. This avoids accepting arbitrary executable model files.

Feature Extraction Design

MediaPipe supplies 21 world landmarks for a detected hand. GestureForge normalises them as follows:

```
origin = wrist
scale = distance(wrist, middle_finger_mcp)
normalised_landmark_i = (landmark_i - origin) / scale
feature_vector = flatten(normalised_landmarks_1_to_20)
```

The wrist landmark is removed after normalisation, leaving 20 landmarks times three coordinates, or 60 features. The preserved orientation helps distinguish different static poses, while translation and approximate hand-camera distance are reduced.

Classifier and Rejection Design

Both classifiers use the same train-only scaling:

```
z = (x - mean_train) / std_train
```

The classifier settings and selected pilot configuration are:

- kNN: KNeighborsClassifier(n_neighbors=5, weights="distance").

- SVM: SVC(C=1, gamma="scale", kernel="rbf").

- Provisional assignment demonstration choice: kNN with rejection multiplier 2.0.

- Model version: prototype-v2.

The app CLI and model API still default to multiplier 1.5; the selected P01 demonstration explicitly uses --multiplier 2, and experiments/pilot.json sets 2.0 for evaluation.

For a predicted class c, the model computes the distance from the scaled live feature vector z to the nearest scaled training example in class c:

```
d(z, c) = min ||z - z_i|| for all training examples z_i with label c
```

For each class, the rejection threshold is the 95th percentile of same-class reference distances, multiplied by the configured multiplier. The original prototype-v1 implementation excluded only the exact sample being scored. In prototype-v2, the evaluation and live training paths supply trial IDs, so reference neighbours exclude the whole originating training trial. Direct model calls without trial IDs retain sample-level exclusion:

```
reference_distance_i = nearest_same_class_distance(training_sample_i, excluding same trial)

threshold_c = quantile_0.95({reference_distance_i}) * multiplier

accept if d(z, c) <= threshold_c
```

This gives kNN and SVM the same rejection layer and makes false-action protection partly independent of the classifier's raw label. SVM decision values are not treated as confidence percentages. The prototype-v2 correction was guided by development validation results, so the reported pilot metrics are not untouched held-out test performance.

Controller State Machine

State	Meaning	Transition trigger
PAUSED	Predictions are ignored	Start camera control calls resume
WAIT_NEUTRAL	System waits for a fresh neutral hold	300 ms accepted NEUTRAL moves to READY
READY	System may consider executable commands	Accepted NEXT starts CANDIDATE
CANDIDATE	One executable command is being held	400 ms stable NEXT emits one command event and moves to LATCHED
LATCHED	A command has already fired for the current hold	Accepted NEUTRAL begins rearm through WAIT_NEUTRAL

Freshness rules reject future, stale, duplicate, out-of-order, and gapped predictions. The maximum accepted prediction age is 200 ms and the maximum frame gap is 200 ms. The assignment executable command set is {"NEXT"}.

Related Work

MediaPipe Hands demonstrates real-time hand landmark tracking from RGB input and supports on-device interaction scenarios. GestureForge uses MediaPipe as an input subsystem, not as the team's own classifier contribution. Its limitation for this project is that good landmarks do not automatically solve deliberate command recognition or false-action control.

Wobbrock, Wilson, and Li's $1 recognizer shows the value of lightweight, example-based gesture recognition for prototypes. GestureForge shares the assignment-friendly emphasis on fast prototyping and personal examples, but the interaction domain differs because GestureForge uses static hand poses from a webcam rather than 2D unistroke paths.

Cover and Hart's nearest-neighbour classification supports the use of instance-based classification for small datasets. The limitation is that nearest-neighbour performance depends strongly on representative training examples and distance choice.

Cortes and Vapnik's support-vector networks support the SVM comparison. The limitation is that an RBF SVM can be less intuitive to explain and requires careful parameter selection.

NASA's lifecycle material supports using gated reviews and maturity checks. The project adapts the idea rather than claiming a full aerospace lifecycle.

Data Collection Plan

The assignment pilot uses one consenting participant. The user has consented to local numeric P01 collection using the right hand, with no raw frames and GitHub exclusion. The collection script records separate train, validation, and validation-negative sessions into data/local/pilot/manifest.json by default when run with the current machine's working camera index:

```
python -m gestureforge.collect --camera 1 --split train --participant P01 --hand Right --setup "desk lighting, plain background" --consent --output data/local/pilot
```

```
python -m gestureforge.collect --camera 1 --split validation --participant P01 --hand Right --setup "desk lighting, plain background" --consent --output data/local/pilot
```

```
python -m gestureforge.collect --camera 1 --split validation_negative --participant P01 --hand Right --setup "desk lighting, plain background" --consent --output data/local/pilot
```

Each standard session uses four labels, eight trials per label by default, a two-second stable pose interval, and five scheduled samples per trial. Validation-negative uses UNKNOWN-labelled records to measure offline classifier acceptance of unenrolled poses. It does not measure live controller actions or an actions-per-minute rate; see the collection protocol.

The completed TRAIN and VALIDATION sessions each contain 32 trials, eight per label, and 160 of 160 scheduled samples were valid. The validation-negative session contains 8 trials and 40 valid scheduled samples from unenrolled poses. Participant identity remains pseudonymous. Consent for private marker inclusion of P01 numeric feature data has been granted and the private package option has been implemented; retention wording still needs to be recorded before final submission. This report does not publish individual feature vectors, trial IDs, thresholds, raw images, raw video, or raw event logs.

Evaluation Plan

Run from the repository root with the retained, consented P01 dataset. The selected-setting config is experiments/pilot.json:

```
{
```

```
"dataset_path": "data/local/pilot/manifest.json",
```

```
"algorithms": ["knn", "svm"],
```

```
"rejection_multiplier": 2.0
```

```
}
```

```
.\.venv\Scripts\python.exe -m gestureforge.evaluate --config experiments/pilot.json --output experiments/local/reproduced-pilot.json
```

The dataset and full result JSON are private local artifacts, absent from the source ZIP and GitHub. Recollecting data produces a new experiment, not a reproduction of the recorded P01 scores. Marker reproduction should use the private P01 package, with retention wording and final package location recorded before submission.

The evaluation reports:

- Trial-level prediction requiring at least three matching labels across five scheduled slots.

- Missing, invalid and rejected slots count as UNKNOWN; a trial without a majority is UNKNOWN.

- Macro-F1 across the four enrolled labels on VALIDATION trials; negative samples are evaluated separately.

- Per-class precision, recall, and F1.

- Confusion matrix including UNKNOWN predictions.

- Per-sample acceptance/rejection records from which counts can be derived, plus the validation-negative accepted command-label fraction for NEXT, PREVIOUS and BOOKMARK. This fraction is not a live action rate.

- Fit and prediction timing.

Synthetic fixtures may be used to test the software path, but the assignment report must label them separately and not present them as participant results.

Pilot aggregate results and local artifact paths are recorded in PILOT_RESULTS.md. The original sample-level leave-one-out baseline reached best macro-F1 0.5667 at multiplier 2.0, where 16 of 32 genuine validation trials scored UNKNOWN. The corrected prototype-v2 whole-trial exclusion raised validation performance substantially:

Multiplier	kNN macro-F1	SVM macro-F1	Validation trials correct	UNKNOWN validation trials	Validation-negative accepted labels
1.0	0.8443	0.8443	25/32	6	25/40 BOOKMARK; 0 NEXT; 0 PREVIOUS
1.5	0.9496	0.9496	30/32	1	25/40 BOOKMARK and 10/40 NEUTRAL; 0 NEXT; 0 PREVIOUS
2.0	0.9686	0.9686	31/32	0	25/40 BOOKMARK and 10/40 NEUTRAL; 0 NEXT; 0 PREVIOUS

At multiplier 2.0, both classifiers produced the same trial-level confusion matrix. Rows are intended labels; columns are predictions.

Intended	NEXT	PREVIOUS	BOOKMARK	NEUTRAL	UNKNOWN
NEXT	8	0	0	0	0
PREVIOUS	0	8	0	0	0
BOOKMARK	0	0	8	0	0
NEUTRAL	0	1	0	7	0

Label	Precision	Recall	F1
NEXT	1.0000	1.0000	1.0000
PREVIOUS	0.8889	1.0000	0.9412
BOOKMARK	1.0000	1.0000	1.0000
NEUTRAL	1.0000	0.8750	0.9333

Classifier	Fit including scaler/rejection (s)	Mean offline prediction per sample (ms)
kNN	0.0121	2.730
SVM	0.0080	0.548

kNN and SVM tied on trial-level validation metrics for this dataset. SVM was faster per saved sample and is a standard nonlinear boundary method, but its decision values are harder to explain to a marker and are not used as confidence percentages. kNN is slower here but still only 2.730 ms per saved feature vector, is simple to rebuild from numeric features without executable model persistence, and is easier to explain for a personal enrolment prototype. Both classifiers rebuild from numeric data without executable model files. These timings exclude camera processing and the controller hold, and do not measure end-to-end latency. The assignment demonstration therefore uses kNN with multiplier 2.0; this is a provisional assignment choice, not a frozen final-project model decision.

The validation-negative results show a serious open-set weakness. For all corrected settings, 25 of 40 negative samples were accepted as BOOKMARK, while zero were accepted as NEXT or PREVIOUS. Because BOOKMARK is recognition-preview-only in the assignment, this does not execute a command in the current prototype, but it means final full-command safety is not met or claimed.

Subsystem Test Plan

The assignment executes tests for implemented prototype subsystems and keeps the final-project checks planned rather than claiming them early.

Subsystem	Assignment test status	Final-project extension
Capture and landmarks	Model asset smoke, blank-image check, live camera checks and P01/P02 qualitative operation	Repeat on final release laptop, include camera unavailable and recovery messages
Feature extraction	Automated tests for 60-feature shape, invalid scale, wrong/multiple hands and non-finite values	Re-run unchanged against final profile workflow
Dataset and manifest	Collection CLI, split integrity and P01 aggregate data summary	Add GUI enrolment data, corrupt/version-incompatible profile handling
Classifiers	kNN/SVM comparison on same features and scaling; macro-F1, confusion and timing reported	Bounded parameter search, frozen configuration, untouched TEST split for two profiles
Rejection	Validation-negative analysis reports accepted unfamiliar poses	Improve BOOKMARK open-set rejection before enabling it as an action
Controller	Unit tests for neutral arm, hold, latch, stale frames, gaps, pause and tracking loss; P01/P02 hold evidence	Formal twenty five-second holds per profile and replay comparison with/without protection
Viewer	Fixed ten-card deck, no wrap, duplicate event rejection and manual fallback	PREVIOUS, BOOKMARK, saved position/bookmarks and validated custom decks
Release	Same-computer source archive/setup checks and P02 assisted qualitative use; guide-only setup, offline/pause/reopen checks and LMS verification remain assignment gates	Repeat acceptance against the final full-system release

Current Verification Evidence

The following combines recorded 23 September hardware checks, the 27 September software verification (60 tests, lint, type checking across application, scripts and tests, GUI smoke and synthetic evaluation passed), and P02 qualitative feedback recorded on 26 September. Software passes do not certify physical-laptop or final-project performance.

Evidence area	Outcome	Limit
Software verification	Ruff, mypy, pytest, GUI smoke launch and synthetic evaluation passed; exact outputs are in verification.json	Synthetic evaluation is software evidence, not participant performance
Setup/package checks	Source archive setup and model checks passed on the development computer; see CLEAN_INSTALL.md, SETUP_CHECK.md and WINDOWS_LAPTOP_CHECK.md	Guide-only second-laptop acceptance is still open
Camera and landmarks	Camera index 1 produced usable non-black video; visible/topmost preview detected a single hand in 338 of 1,145 frames and closed cleanly	Detector activity is not gesture-classification accuracy
P01 data	TRAIN and VALIDATION each completed 32 trials with 160/160 valid scheduled samples; validation-negative completed 8 trials with 40 valid samples	One consenting right-hand profile only
P01 classifier comparison	kNN and SVM both reached macro-F1 0.9686 at multiplier 2.0; selected-setting tables are included above	Development validation only, not untouched held-out testing
Open-set rejection	Validation-negative accepted 0/40 as NEXT and 0/40 as PREVIOUS	25/40 were accepted as BOOKMARK, so full-command safety is not claimed
Live NEXT path	Local event log records genuine gesture NEXT events advancing viewer positions 1, 2 and 3 through webcam, landmarks, kNN, controller and viewer	Formal latency, false-action and twenty-hold runs are final-project checks

Evidence area	Outcome	Limit
Hold/rearm behaviour	P01 confirmed two sustained NEXT holds each advanced exactly once after NEUTRAL rearm; P02 reported a five-second hold fired once	P02 result is participant-reported and not quantitative
P02 laptop feedback	P02_LAPTOP_TEST.md records assisted ZIP use, P02 right-hand data, Card 1 to Card 10 NEXT traversal, NEUTRAL rearm and preview-only labels	Exact package/environment, guide-only setup, pause/reopen and disconnected-network checks remain open
Review status	REVIEW.md records a bounded review	Tutor approval, final human report review and signed declaration are not complete

Remaining Assignment Gates

Track completion in the assignment checklist and next steps. The generated archive contains source and draft documentation, not the complete reviewed submission.

- Human review of the exported report draft; regenerate after any content changes.

- Complete remaining P02 laptop checks: exact package/environment and test date, guide-only setup, pause/restart, camera release/reopen and disconnected-network use. Retain the assisted-use observations separately from complete release acceptance.

- Tutor feedback on scope, marking weights, evidence format and AI-use questions in TUTOR_REVIEW.md.

- Private marker inclusion of P01 numeric feature data is permitted and the private package option has been implemented; record retention wording and final package location before upload.

- Team registration/tutorial membership, full identifiers and verified contributions in CONTRIBUTIONS.md, plus unit AI-use guidance and review of AI_PROMPT_APPENDIX.md.

- Signed group declaration.

- LMS upload and download verification.

Mandatory Development Tools

The assignment sheet says the submission will not be marked if project management, version control, collaboration and communication tools are not reported. Current evidence is:

Required category	Actual use	Remaining gap
Project management	Agile backlog in backlog.csv, rendered sprint board, assignment/final checklists and issue-style acceptance criteria	Tutor updates and final sprint review need human completion notes
Version control	Git repository under code, private GitHub repository on main, redundant Codex branch removed after merge	Include the repository URL or source archive in the final submission as allowed
Collaboration	GitHub repository/project board and contribution record in CONTRIBUTIONS.md	Jameson and Deston are the supplied names; full identifiers, verified responsibilities and agreed contribution percentages remain pending
Communication	WhatsApp, confirmed by the user on 27 September; tutor questions are tracked in TUTOR_REVIEW.md	Actual team activity summaries and tutor replies remain pending; no meeting or message content is invented

Software Tools Used

Tool	Actual project use
Python 3.11.16	Local runtime
PySide6	Desktop GUI
OpenCV	Camera capture and preview frame handling
MediaPipe Hand Landmarker	Pretrained hand landmark detection
NumPy	Feature matrices and numeric operations
scikit-learn	StandardScaler, kNN, SVM, and metrics support
pytest	Automated regression tests
Ruff	Python linting
mypy	Static type checking
Git	Local version control
GitHub	Private repository, milestones, issues, and project board
WhatsApp	Team communication confirmed by the user; activity details pending
OpenAI Codex	Planning, code generation, tests, documentation, and review support under human responsibility

Risks

Risk	Impact	Mitigation
Camera index or backend opens the wrong device	Live demo or collection may silently use unusable video	Verify the device index with python scripts/check_environment.py --camera --camera-index 1 before collection; recheck after unplugging or replugging cameras
Open-set rejection accepts BOOKMARK on negative poses	Final full-command project would risk unintended bookmark actions	Keep BOOKMARK preview-only for the assignment; improve rejection before final full-command release
Formal twenty-hold duplicate run is not complete	The final duplicate-action target needs more than the smoke check	Treat the two-hold result as smoke evidence only; run the formal series before claiming the final target
Report writing or final review is rushed	Evidence may be omitted or overstated	Use the checklist and audit to reconcile every claim against recorded evidence
Overscoped assignment	Late or incomplete submission	Keep assignment to NEXT execution only
Synthetic results are confused with participant results	Misleading evaluation claims	Label synthetic software tests clearly
Missing declaration or upload verification	Marking penalty	Keep submission checklist and human gates visible

References

- Zhang, F. et al. (2020). MediaPipe Hands: On-device Real-time Hand Tracking. https://arxiv.org/abs/2006.10214

- Wobbrock, J. O., Wilson, A. D., and Li, Y. (2007). Gestures without libraries, toolkits or training: a $1 recognizer for user interface prototypes. https://depts.washington.edu/acelab/proj/dollar/index.html

- Cover, T. and Hart, P. (1967). Nearest Neighbor Pattern Classification. https://isl.stanford.edu/~cover/papers/transIT/0021cove.pdf

- Cortes, C. and Vapnik, V. (1995). Support-vector networks. https://doi.org/10.1007/BF00994018

- NASA. 3.0 NASA Program/Project Life Cycle. https://www.nasa.gov/reference/3-0-nasa-program-project-life-cycle/

AI Assistance Declaration Draft

AI assistance was used to plan scope, implement prototype code, write automated tests, draft documentation, identify risks, and prepare report scaffolding. The team remains responsible for checking the code, running and preserving verification evidence, validating citations, collecting consented data, and ensuring every submitted claim is accurate. The recovered user-message appendix is in AI_PROMPT_APPENDIX.md; the original planning requests and full implementation-plan prompt have now been recovered for this task. The user confirmed Codex use, but coverage of other Codex conversations, including the assisted P02 setup, remains unconfirmed. The log preserves user wording and separates original prompts from the summarised usage index. Unit-specific AI guidance and human review remain pending.

Appendix A: AI assistance record

Updated 28 September 2026. This record describes actual AI assistance and verification. It does not claim human signoff or compliance with unit-specific guidance that has not been supplied. The user confirmed **Codex** as the AI tool used so far.

Declaration for human review

OpenAI Codex was used substantially for scope planning, implementation, automated tests, troubleshooting, documentation, repository/agile setup and review. AI-generated work includes prototype source code, helper scripts, tests and report text. MediaPipe supplies the pretrained hand detector and landmarks; scikit-learn supplies the classifier implementations. These are third-party components, not team-authored algorithms. GestureForge fits the supplied classifiers to the personal numeric training data.

The team remains responsible for understanding and checking the submitted work, attribution, participant permissions and accurate reporting. Human names, contributions and signoff must be recorded in CONTRIBUTIONS.md. AI assistance is not a student team member and must not be assigned a student contribution percentage.

This is a factual draft for team review. The unit information guide and required AI citation format remain to be checked. No specific model/version has been inferred from the model selected today.

Prompt record

The user prompt log replaces the earlier representative excerpts. It preserves stable prompt IDs, recorded UTC timestamps where available, original wording, the full implementation-plan prompt, and clarification questions with the user's answers. Original-message hashes support checking against the retained source. Images, internal system instructions, hidden reasoning and tool outputs are not part of this user-prompt export.

Prompt IDs	Request or clarification	Related work and evidence
PR-001 to PR-006	Assignment/final split, implementation, agile tools and repository	Build plan, agile record, source and tests
PR-007 to PR-017	Camera troubleshooting, P01 consent and live hold/rearm observations	Hardware record, pilot results
PR-018 to PR-020	Main branch, README and another Windows laptop	GitHub record, Windows guide
PR-021 to PR-025	P02 feedback, private P01 sharing, local operation and gesture scope	P02 evidence, user guide, private package check
PR-026 to PR-027	Continue packaging and tighten records without assumptions	Report, decision log, submission inventory
PR-028 to PR-030	Confirm names/tools and pending team, retention and laptop facts	Contributions, checklist
PR-031	Audit the repository and remove redundant code/files	Cleanup audit; unused dataset helpers and duplicate controller guard removed; generated artifacts cleaned

This index identifies related artifacts, not proof that every requested result is complete. Each artifact states its evidence status.

Verification of AI assisted work

Work	Verification recorded	Limit
Source and tests	28 September verification: 60 tests, Ruff, mypy, GUI smoke and synthetic comparison passed	Automated checks do not replace human understanding or live hardware tests
AI comparison	Private archive reproduced P01 kNN/SVM macro-F1 0.9686	Development validation, not held-out generalisation; 25/40 negative samples accepted as BOOKMARK
Live prototype	P01 participant-observed hold/rearm check and P02 assisted laptop report	Qualitative smoke evidence; final-project performance targets remain untested
Documentation	Checked against assessment files, code and recorded evidence	Human approval, administrative facts and external AI-history coverage remain pending

Coverage and remaining review

Original planning and implementation messages have now been recovered for this task. That closes the earlier missing-plan gap; it does not establish completeness across all Codex chats. Other conversations, including P02's assisted setup, have not been supplied or confirmed complete. The user's statement “codex so far” identifies the tool, not the number of conversations or transcript coverage.

Before submission, confirm all AI conversations used are represented, check unit-specific declaration/citation guidance, review privacy and attribution, and have both team members approve the text. The course specification requires prompts in the report; the report PDF includes this record and the recovered prompt appendix. Do not mark external-history coverage complete until confirmed.

Appendix B: Recovered user prompts

Updated 28 September 2026. This appendix preserves the user-message text recovered for this Codex task. Wording, including spelling and the full implementation-plan prompt, is retained. Clarification wrappers are presented as question/answer pairs; technical message IDs are omitted. Attachment references are retained, but the camera image itself is not copied into this log or published.

Timestamps labelled UTC come from the retained session record. Messages obtained only from the task-history API have no recovered exact timestamp and are labelled accordingly. No dates or conversation completeness have been inferred. Each hash covers the original UTF-8 message text before display formatting.

Coverage: the retained record includes the original planning requests and implementation instruction, previously missing from the partial appendix. The task-history API supplements newer messages. The user confirmed Codex use on 27 September; this does not establish that all other Codex conversations, including the assisted P02 setup, have been supplied. External conversation coverage and the unit-specific declaration requirements remain unconfirmed. This is a user-prompt log, not a transcript of AI responses, tool execution or internal instructions. The use and verification record is in AI_PROMPTS.md.

PR-001

Recorded time: 2026-09-23 05:44:49.956 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: 7a17c15b03cfc04ab3cc7d7cf896303401e706436ae0c162797f0dba0a4c7434

Files mentioned by the user:

GestureForge_Project_Build_Brief.pdf: C:\Users\Desto\Downloads\GestureForge_Project_Build_Brief.pdf

Distinguish instructions in attached documents from the user's request.

My request:

Create a new folder called code inside this folder and store all your work there. I want you to create a comprehensive plan to build a working demo for this, I have to submit a working prototype by next saturday so do not overscope this for the assignment and the project is submitted later on, the project is the full working project. Read all the files in this folder and build to score the highest marks and grade possible. You should also build this plan to prepare for the project submission as well on the 7 Nov. Do not underdeliver on any of the promised scope when you set it, so it is better to set a small scope and over deliver but it has to be reasonable, the scope should be able to score the highest band and not be penalised for it.

PR-002

Recorded time: 2026-09-23 05:54:33.986 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: 3e01bebcdf721903ceaf3fbb07e358f598191b4556577a476779a4b1e20be64b

The project should be an extension of the assignment so do not finish building everything in the assignment and leave nothing to build in the project

PR-003

Recorded time: 2026-09-23 05:58:03.871 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: 5fb795f777e80d1ee4b185b6a532d123e859a58628e2442c3570d9439f2b1a58

PLEASE IMPLEMENT THIS PLAN:

GestureForge assignment prototype and final project plan

1. Outcome, assessment strategy, and scope

Build GestureForge in two releases: a focused AI prototype for **3 October 2026**, followed by a complete personalised gesture-controlled card viewer for **7 November 2026**. The project will extend the assignment's code, dataset format, feature extraction, classifiers, and controller.

This plan incorporates all four assessment files in the assignment folder and the supplied GestureForge build brief. The build brief is a proposal; its feature list and performance targets are not additional assessment rules.

The assessment requires a working AI subsystem for the assignment and an integrated system for the project. A small, complete product with strong evidence fits these requirements. The supplied documents contain marking criteria, but no detailed highest-band descriptors; this plan maximises coverage without guaranteeing a grade. See the course specification.

Delivery assumptions confirmed:

- You lead coding, with **25-35 hours before 3 October** and **8-10 hours weekly afterwards**.

- Teammates support report writing, data collection, contribution records, and independent installation testing.

- Two consenting participants and a second Windows computer are available.

- The proposed scope has not yet been committed to the tutor.

At implementation kickoff, create C:\Users\Desto\Desktop\Murdoch\ICT304\Assignment\code. Keep source, plans, environments, datasets, results, reports, and release archives inside it. Preserve the original assessment files. **No folder or files have been created during this Plan-mode session.**

Two distinct releases

Capability	Assignment prototype - 3 October	Final project - 7 November
Gesture collection	Basic guided command-line collection for one participant	In-app enrolment for multiple local profiles
AI subsystem	Four labels: NEXT, PREVIOUS, BOOKMARK, NEUTRAL; implement and compare kNN/SVM	Refined selection and rejection; evaluate two personal profiles
Live operation	Webcam preview, prediction label, controller state	Complete practice and control workflow with actionable feedback
Actual viewer control	**NEXT only**, on a fixed supplied deck	NEXT, PREVIOUS, BOOKMARK
Activation protection	Neutral-to-arm, hold-to-trigger, one action per hold, pause, tracking-loss reset	Extend protection to profile/model switches, recovery, and all commands
Persistence	Save feature data, configurations, and experiment outputs	Profile creation/reload/deletion, deck position and bookmarks
Deck support	One supplied text-card deck	Supplied deck plus validated JSON deck loading
Evaluation	Preliminary classifier comparison and prototype tests	Held-out evaluation, all-subsystem tests, live tasks, stability and release checks
Submission	Design report, runnable prototype, initial results, declaration	Complete system, final report, guide, slides, demo video, declaration

PREVIOUS and BOOKMARK are recognised labels in the prototype, visibly marked **"recognition preview only"**. They do not execute viewer actions until the project release.

Final scope exclusions: dynamic gestures, two-handed commands, sign-language translation, PowerPoint/OS control, cloud accounts, mobile apps, content generation, deep-network training, and an executable installer. These are future possibilities, not promised features.

2. Implementation and interfaces

Technology and organisation

Use **Python 3.11, PySide6 Widgets, OpenCV, MediaPipe Hand Landmarker, NumPy, scikit-learn, and pytest**. Begin with a compatibility and webcam check on both Windows computers, then pin the versions that actually pass. Do not assume the combination has already been tested.

Python 3.11 falls within the documented Python requirements for MediaPipe and Qt for Python. Package compatibility and camera operation still require practical verification.

Use one repository under code, organised into application source, tests, data, assets, experiments, documentation, and release outputs. Save this plan as code/docs/BUILD_PLAN.md when implementation begins.

Provide these stable entry points:

- python -m gestureforge - launch the desktop application.

- python -m gestureforge.collect - collect labelled trials.

- python -m gestureforge.evaluate --config ... - reproduce classifier comparisons.

- python -m pytest - run automated tests.

Use a simple application window for the assignment: camera preview, current prediction, controller status, current card, and Start/Pause/Next controls. Extend it with Setup, Practice, and Viewer sections for the project. Do not build a separate disposable prototype.

Processing pipeline

Webcam -> landmarks -> normalised features -> classifier/rejection -> activation controller -> viewer -> event log

MediaPipe supplies pretrained hand landmarks; the team's AI contribution is the personal classifiers, preprocessing, rejection, comparison, and application integration. Its documented output includes 21 landmarks and handedness. Hand Landmarker documentation.

Use a worker for camera capture and synchronous video-mode inference, keeping widget updates on the UI thread. Permit only the latest pending display/result update; do not accumulate frame queues. Stop and release the camera cleanly.

Shared contracts

Interface	Required content
Observation	Session/generation ID, profile ID, frame ID, monotonic capture time, hand count, handedness, landmarks, validity reason
Prediction	Observation identity, label, accepted/rejected status, rejection reason, model/configuration version
Command event	Unique event ID, originating prediction, command, execution time, resulting card position
Dataset manifest	Participant, session, trial, intended label, split, sample timestamps, validity, hand, collection conditions
Profile	Schema/feature version, chosen hand, training-data reference, model configuration, rejection settings, command mapping
Deck	Schema version, deck ID, title, ordered cards with unique IDs, titles and plain-text bodies

Keep these contracts unchanged between submissions except for explicitly versioned additions.

AI implementation

- Enrol **one chosen hand per profile** and detect up to two hands. Reject no hand, multiple detected hands, wrong hand, non-finite landmarks, and degenerate scale.

- Subtract wrist position from world landmarks, divide by wrist-to-middle-knuckle distance, remove the wrist triplet, and flatten to **60 features**.

- Preserve pose orientation. Mirror the preview only; use the same unmirrored input convention for collection and inference.

- Fit a StandardScaler using training data only. Apply the identical feature/scaling policy to both classifiers.

- Prototype defaults: kNN with k=5, distance weighting; RBF SVM with C=1, gamma=scale, and probability estimation disabled.

- Final bounded search: kNN k□{3,5,7} with uniform/distance weighting; SVM C□{0.1,1,10} with gamma□{scale,0.1}.

- Add a shared rejection rule based on distance to training examples of the predicted class. Derive class-specific reference distances from training data; select a multiplier from {1,1.5,2} using validation positives and negatives.

- Select configurations using validation macro-F1 subject to the false-action target. Break ties using fewer false actions, then lower inference time; prefer kNN if effectively equivalent. If neither meets the target, report the shortfall and improve development data before freezing evaluation.

Pipelines help prevent preprocessing leakage. SVM decision values must not be labelled as confidence percentages. scikit-learn leakage guidance, SVC documentation.

Save numeric training data and JSON configurations; rebuild these small classifiers on profile load. This avoids accepting arbitrary pickle/joblib files, whose loading can execute code. Record versions and checksums for reproducibility. Model persistence guidance.

Controller and viewer behaviour

Implement a separately testable state machine: **PAUSED -> WAIT_NEUTRAL -> READY -> CANDIDATE -> LATCHED**.

- Start with a 300 ms neutral hold and 400 ms command hold.

- Reject observations older than 200 ms, out-of-order timestamps, and gaps exceeding 200 ms.

- Execute once per continuous command hold; require neutral before another command.

- A changed or unknown candidate cancels the pending hold. Tracking loss requires neutral again.

- Pause, camera failure, and profile/model changes invalidate pending generations.

- Enrolment and practice cannot dispatch commands.

- The viewer ignores duplicate event IDs, does not wrap at deck boundaries, and sets bookmarks idempotently.

- Retain keyboard/mouse controls and label their events separately.

The project adds validated custom decks, atomic profile/state writes, reload/delete operations, and clear recovery messages. Raw recording stays off during normal use.

3. Evidence and assessment coverage

Assignment priorities

The assignment feedback sheet assigns requirements/prototype **30%/20%**; the course PDF assigns **35%/15%**. Ask the tutor to confirm, while satisfying both.

Criterion	Evidence to deliver
Systems engineering - 10%	Applied lifecycle, review gates, decisions, risks, tutor feedback, and requirement changes
Requirements/problem/tests - 30-35%	Defined user and task, bounded operating conditions, numbered requirements, subsystem contracts, data needs, full-system test plan
Technique investigation - 20%	Explain both algorithms, investigate trade-offs, run a fair preliminary comparison, justify the provisional choice
Prototype/results - 15-20%	Runnable webcam AI subsystem, actual NEXT action, measured results, reproducible commands
Innovation - 10%	Personal enrolment combined with deliberate command activation; demonstrate the contribution without claiming new hand tracking
Writing/references - 10%	Clear diagrams, consistent terminology, accurate attribution, source references, complete submission

Prepare the assignment report alongside implementation: problem and user workflow; related approaches; lifecycle; requirements; architecture; data; technique comparison; prototype/results; full test plan; milestones; risks; tools and contributions; references and AI-use appendix.

Investigate at least three relevant primary sources covering hand tracking, gesture interaction, and personalised/lightweight recognition. Explain their relevance and limitations; do not use software documentation alone as the related-work discussion.

Data and evaluation

Assignment pilot

- One consenting participant.
- Separate TRAIN and VALIDATION recording sessions.
- Eight trials for each of four labels per session.
- Each trial has a two-second stable pose and release interval; sample five predetermined timestamps.
- Maximum **320 feature vectors across 64 trials**, plus a short separate validation-negative sequence.
- Report validation results as preliminary development evidence, not final generalisation results.

Final evaluation

- Two participants, each with their own trained profile.
- Separate TRAIN, VALIDATION, and untouched TEST sessions.
- Twelve trials per label per session, up to five vectors per trial: **288 trials and 1,440 vectors maximum**.
- Keep all frames from a trial in one split; never randomly split neighbouring frames.
- Freeze configuration and split manifests before opening final test results.
- Retain invalid detections in the manifest and count them as UNKNOWN when scoring.

For trial-level comparison, use the majority of the five scheduled predictions; no majority becomes UNKNOWN. Report macro-F1 over the four enrolled labels, per-class precision/recall, and a confusion matrix including UNKNOWN predictions. Rejected genuine gestures count as misses.

Report each participant separately. Personal-profile results do not establish performance on unseen users without enrolment.

For the final system, measure:

- A prescribed ten-card navigation/bookmark task.
- Twenty five-second command holds per profile: **zero duplicate actions**.

- Ten minutes of non-command activity per profile: target **at most one unintended action**.

- Recognition target: **macro-F1 ≥0.90 per profile**.

- Action latency from annotated stable-pose onset: target **95th percentile below one second**, including the hold.

- Fit time: target **below 60 seconds per candidate fit**; report parameter-search time separately.

- Thirty-minute stability run, observation rate, peak process memory, and clean camera shutdown.

These are engineering targets, not already achieved results. Review them after the pilot and baseline them before final testing; never lower targets retrospectively to conceal failure.

Use one offline replay experiment to compare the same predictions with and without hold/rearm protection. Label replay evidence separately from live webcam testing.

Required tests and release evidence

Area	Essential scenarios
Features/data	Correct shape and normalisation; invalid scale; wrong/multiple hands; split integrity; no training/test leakage
Classifiers	Both techniques reproduce results from saved inputs/configuration; rejected poses count correctly
Controller	Long hold, neutral rearm, changed pose, occlusion/re-entry, stale and duplicate events, pause
Viewer	First/last card, exactly-once actions, bookmark persistence, keyboard fallback
Profiles/decks	Round-trip save/load, deletion, corrupt/version-incompatible files, malformed deck
Integration	Real camera-to-action path; commands disabled during enrolment; no old actions after switching profiles
Release	Fresh installation on second computer, offline operation after setup, missing model asset, camera unavailable, stability

For the assignment, execute tests for implemented parts and provide planned tests for the remaining final-system requirements. For the project, execute and report tests for **every subsystem**.

The final project feedback sheet weights requirements 20%, design 35%, evaluation 20%, guide/code 15%, and presentation quality 10%. It also identifies deductions for missing demo video and incomplete subsystem testing. Protect those deliverables before adding polish.

4. Schedule, ownership, and release gates

Assignment: 23 September-3 October

Dates	Work and exit gate	Your budget
23-24 Sep	Create repository/layout; verify dependencies, camera and landmarks; draft scope/requirements; prepare tutor review	4 h
25-26 Sep	Implement collection, feature pipeline, manifest and pilot data; train both defaults	6 h
27-28 Sep	Build minimal viewer and controller; join webcam prediction to one real NEXT action	6 h
29-30 Sep	Preliminary comparison, rejection tuning, targeted tests, measured outputs	5 h
1 Oct	Freeze prototype; independently install; reconcile report claims with code/results	4 h
2 Oct	Final package review, declaration, submit early, download and verify uploaded files	3 h
3 Oct	Reserved correction window before the official midnight deadline	Up to 5 h

Total: approximately 28 hours plus five hours contingency.

Teammates draft the report while development proceeds, maintain source/contribution records, cross-check labels, and perform the second-computer installation. You review technical explanations and demonstrate that you

understand the code.

Assignment completion means a real webcam gesture advances a card once, both classifiers produce reproducible initial results, the prototype installs independently, and the report accurately distinguishes implemented and planned functionality.

Project: 4 October-7 November

Dates	Extension work and exit gate	Your budget
4-11 Oct	Add guided enrolment/practice, profile create/reload/delete, configuration validation	9 h
12-18 Oct	Add PREVIOUS/BOOKMARK, JSON decks, saved state, profile-switch protection and recovery; **feature freeze**	10 h
19-25 Oct	Complete validation/search, freeze candidates, collect held-out data, run live tasks and controller replay	9 h
26 Oct-1 Nov	Fix defects, rerun affected tests, second-machine release test, stability run; **release candidate**	9 h
2-6 Nov	Final report/guide review, demo video, slides, presentation rehearsal, submission verification	8 h
7 Nov	Deadline buffer only	Reserve

This leaves approximately **45 further developer hours**, with teammates owning substantial documentation and evaluation support.

Adapt NASA's lifecycle into real project gates: concept/feasibility and preliminary design for the assignment; detailed design, integration and verification for the project; operation, maintenance and closeout plans for handover. Do not claim long-term operation has already happened. NASA lifecycle reference.

Schedule protection: optional styling and extra charts are removed first. Do not cut the AI comparison, working camera path, controller correctness, reproducibility, or required submissions. Any change to committed final scope must be recorded and discussed with the tutor before it becomes an omission.

5. Submission and completion controls

Use Git/GitHub for version control, GitHub Issues/Projects for task management, pull requests/shared report drafts for collaboration, and the team's existing chat plus weekly tutor meetings for communication. Keep genuine usage evidence and back up each release.

At kickoff, confirm team registration, same-tutorial membership, requirements approval, marking-weight discrepancy, and the unit's full AI-use guidance. Record actual names and responsibilities without inventing approvals, signatures, contributions, or research results.

Maintain:

- A requirement-to-test-to-evidence matrix with assignment/project status.

- A decision/risk log and weekly tutor-update record.

- Dependency/model sources, licences, versions, and checksums.

- Participant permissions, pseudonymous data, permitted marker access, and an agreed retention arrangement.

- An AI-assistance declaration and prompt appendix from the start.

- A contribution record matching actual work and the Group Declaration Sheet.

Assignment package: formal design report, prototype source, permitted pilot data, model/configuration assets, initial results, installation/run instructions, and signed group declaration.

Final package: integrated source and assets, permitted reproducible datasets and configurations, final report, complete user/developer guide, test evidence, presentation slides, demo video, AI-use appendix, and signed declaration. Prepare the specified **ten-minute presentation**; the supplied files do not specify a demo-video duration.

The final demo must show a user enrolling gestures, validating them, controlling a deck with all three commands, holding without repeated actions, pausing safely, and reloading saved state. Show actual results and one limitation.

Before each submission, install and run the release archive on the second computer using only its instructions. After upload, download the LMS package and check it again. Missing tool reporting or declaration can prevent marking, and non-working code/demo carries a stated 50% deduction.

Completion means every committed capability is implemented, every subsystem has recorded verification, the marker can reproduce the results, and both team members can explain their contributions. Additional features are unnecessary.

PR-004

Recorded time: 2026-09-23 05:59:10.694 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: 654c0fbaf20198455efd4bca12b7e66dc2dbcaf2204e02d56c1264032dc06a6f

remember to use agile development tools as well

PR-005

Recorded time: 2026-09-23 06:03:40.631 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: cd5ab6ea9052cda89890434bc03cf466e48cd1236a5873934dda2e02f9756acc

Question: Do you already have a GitHub repository or project board for GestureForge? Send its URL if so. I'm setting up Git and a local sprint board now so work can continue either way.

User answer: No I do not have, create a new repo for me

PR-006

Recorded time: 2026-09-23 06:04:55.503 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: 2aac1e79010e9b6ea12c7597e646d0f2bcb7aab0c59ab681ab7900e16e84558b

ensure all requirements are met in order to score the highest band and not be penalised

PR-007

Recorded time: 2026-09-23 07:04:15.000 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: b1a02d542da38e1b1fedb743c4ee8612d44052760d9468f3608c527fdc009474

Question: The default webcam returned no image, and the alternate Windows backend stalled. Is a physical webcam connected and enabled on this computer, or will you demonstrate on another laptop?

User answer: No webcam connected yet

PR-008

Recorded time: 2026-09-23 07:06:13.658 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: ce71699de805779700c06afb106948114e71f808fc5aca20a82ffef57ead334b

I can connect a webcam whenever, should I do it now or when you finish this run

PR-009

Recorded time: 2026-09-23 07:07:39.947 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: 9defb9fb3f5fdc28a5747dcde561beeefcb8478a5e2d5e6976b9c110d48ba1b8

Question: Is the webcam connected and ready for me to test?

User answer: Connected and ready

PR-010

Recorded time: 2026-09-23 07:14:48.194 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: 9c607e2c435b40a2d7eebcd30ae10f19e4ef9fd85fcc2d866c6610b8e295536d

Question: Is the webcam lens uncovered, with your hand clearly in front of it and enough room lighting?

User answer: Yes, uncovered and hand in view

PR-011

Recorded time: 2026-09-23 07:17:25.157 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: 164ed6fcfe4a1b6a7c54a89f2e4fef8f13e24ec18e8c0428284ec978caf1b628

i have changed my webcam, is this one better or should i use the previous one

PR-012

Recorded time: 2026-09-23 07:22:20.741 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: 35340346eef8dd11fb4132a3100c23418ffc4d84824856dc41f2961e40854259

Question: In the Windows Camera app I just opened, do you see a clear moving image, a black preview, or an error? If there is an error, paste its code.

User answer: Clear moving image

PR-013

Recorded time: 2026-09-23 07:22:22.039 UTC. Source: Retained task session record. Type: Attachment reference.

Original-message SHA256: 2688b8debf32c393ae30edeffd3f3db0075366de316f424dd9bb83c68609990d

Files mentioned by the user:

codex-clipboard-418fbe73-2215-43dd-bab0-7133ec3780fd.png: C:/Users/Desto/AppData/Local/Temp/codex-clipboard-418fbe73-2215-43dd-bab0-7133ec3780fd.png

Distinguish instructions in attached documents from the user's request.

My request:

<image name=[Image #1]
path="C:\Users\Desto\AppData\Local\Temp\codex-clipboard-418fbe73-2215-43dd-bab0-7133ec3780fd.png">

</image>

PR-014

Recorded time: 2026-09-23 07:33:16.221 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: 9d56aa590910273ab662a37ea169444bca30344dd5332365620d740911f0d526

Question: Did the GestureForge camera-test window show your hand clearly, or was it showing a different view? It processed video but detected no hands.

User answer: I did not see the test window

PR-015

Recorded time: 2026-09-23 07:35:58.689 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: 6b288bd7754c3186ddce6636bf844210616ef28171daf9362815ff1d87a9820b

Question: May I save local numeric gesture samples under pseudonym P01 for your assignment pilot, and which hand will you use? No raw video or images will be saved, and the samples will stay out of GitHub.

User answer: Yes, use my right hand

PR-016

Recorded time: 2026-09-23 07:58:37.386 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: 7ceeed8c688056d908841c9e596cfd82c516811ba3884cdb35e75b73c97d4e2a

Question: In the GestureForge window, did each fist hold advance exactly one card, with another advance only after returning to open palm?

User answer: I need to try again I did not notice it

PR-017

Recorded time: 2026-09-23 08:04:05.588 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: ff3a90bf6448a3caae107d4997d312df130f80dceaa6454c6693f9d55fc9914a

Question: After this retest, did the five-second fist hold move only to card 2, and the next fist hold after open-palm rearming move only to card 3?

User answer: Yes, both holds advanced exactly once

PR-018

Recorded time: 2026-09-23 08:18:31.777 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: 4932684db1a3084d6696da52897cbb771b00bdb31ff4c00dfee4eb7bde1bdf27

 What is left for the assignment build and what is next? Also, I want the branch to be on main and I want my readme to be professionally designed like my memepet repo but make it suitable for assignment submission. Also, resolve any issues that can be resolved for me

PR-019

Recorded time: 2026-09-23 08:40:52.710 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: 13dff62a9637587de21a2aa287197e4fb647a44f8254638cb3870f73b5a2b831

I want you to remove or merge the codex branch into main and ensure that this proiect can be run and used on a separate windows laptop

PR-020

Recorded time: 2026-09-23 08:43:34.313 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: ed0263eea0cf1347b7dc9cf2ff52397e511f5a52e8248b7a6ad8b6f4dcf73416

Question: Is the second Windows laptop available now, and is it Windows 10/11 with an Intel/AMD processor or Windows on ARM? I'm preparing the transferable setup package while checking what I can here.

User answer: Not available yet - Intel/AMD Windows

PR-021

Recorded time: 2026-09-26 08:46:19.836 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: ed0e3ca114b22ea2ecfe673ff1fd1966a4aafbb314f530735a2e8e961f301ea4

This was my friends output after trying it on his laptop:

I tested GestureForge using my own P02 right-hand dataset and laptop camera. Gesture recognition worked best when my hand faced the camera, remained still, stayed at a suitable distance and was positioned similarly to the training samples. Recognition became unreliable when my hand moved, tilted, turned backwards, shifted too far to the left or right, or moved too far away from the camera. When recognition was lost, I usually needed to remove my hand, show NEUTRAL again and then repeat the command gesture.

The NEXT gesture successfully advanced the fixed demo deck from Card 1 to Card 10. Holding NEXT for five seconds produced only one action rather than repeatedly advancing the cards. The system required the NEUTRAL gesture before it would accept another NEXT command. PREVIOUS and BOOKMARK could be recognised and displayed, but they did not perform card actions because they are preview-only features in the current Assignment 1 prototype.

The test confirms that the main webcam-to-NEXT workflow works successfully. However, recognition is currently limited by hand movement, orientation, distance, position and the enrolled right-hand dataset. Future development should include training samples from more positions, distances and orientations, improve rejection testing, and implement the PREVIOUS and BOOKMARK actions.

Is this project ready for assignment submission, if no what is left to complete? Optimise it so it can score the highest grade.

PR-022

Recorded time: 2026-09-26 08:48:46.828 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: 673170933672f79e2d5d47e90844d8f98079f3568e5564ada35a7d9c66202133

Question: To finish the submission details, send any confirmed team names/student IDs and tutor feedback, and say whether you permit your P01 numeric dataset to be included privately for the marker. If those are undecided, I'll leave them clearly pending.

User answer: What is the P01 dataset?

PR-023

Recorded time: 2026-09-26 08:50:16.350 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: a1e960f6c9b3d1faa1f947eb22c10ec6147651dce7f1478cfc08919a46fc34bd

Question: For your friend's test record, do you know the test date, Windows version and project version used, and whether they installed from the ZIP using only the guide? Please also confirm if pause/restart and use with Wi-Fi disconnected were tested.

User answer: They tested with the zip file through codex. The pause/restart and use with WI-Fi disconnected were not tested

Question: May I include your P01 numeric gesture samples in the private assignment package for your tutor/marker to reproduce the results? They will remain excluded from GitHub, and no photos or video will be included.

User answer: Is the samples needed for submission? If yes, include them

PR-024

Recorded time: 2026-09-26 08:50:59.961 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: acc12d143ee1936868e36f80f47e0471edd725eb1e8322ce9d87f9160ff39024

Give me a guide on how I can run this program locally and is it connected to my microsoft powerpoint?

PR-025

Recorded time: 2026-09-26 08:52:18.634 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: 57b3ef32fe1ac91daf9fce71dd7480b702bac37bdcc419dbf16524731a1f617c

Does the other gestures work and is there anything that needs to be fixed?

PR-026

Recorded time: 2026-09-27 08:43:57.564 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: abd98fbbaa8d260373c81e88bc10c740f0c5b0af12c8a333f391423290415667

Continue building from where you left off, and also do I need to connect to my webcam for more training data or is this enough for my current Assignment submission

PR-027

Recorded time: 2026-09-27 12:04:41.760 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: eeee70da2eeeedc619f65288cda383808016d84cde660025595babc9dcd035fc

Tighten my prompt log and my decision log and prepare all the documentation needed for submission, clarify any questions if needed, do not make any assumptions

PR-028

Recorded time: 2026-09-27 12:07:28.705 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: 6982f516a0b9a22f09fa6893e427b79b749f2f556b8c6e553e48bbcdf06a20e1

Question: Please provide the confirmed team member names, student IDs, actual contributions, and collaboration/communication tools used. Also include any tutor feedback or approvals; explicitly say “not confirmed” for anything still undecided.

User answer: Jameson and Deston. Github and Whatsapp so far

PR-029

Recorded time: 2026-09-27 12:09:08.654 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: 00a681e8fcd0ffe2fa0cd1037d4e55f9922605e8540d2b3cc8560bb814764c78

Question: Were any AI conversations or tools used for this assignment outside this Codex chat-including your friend’s laptop setup? List them and say whether you have their prompt transcripts. I need this to establish the prompt log’s coverage without guessing.

User answer: codex so far, tidy this prompt log's as well

Question: For Jameson and Deston, please provide surnames, student IDs, university emails, who did which work, and the contribution percentages you both agree to (total 100%). Also confirm your group name, same tutorial group/team registration, and any tutor feedback. You can mark anything not yet confirmed as pending.

User answer: pending first

PR-030

Recorded time: 2026-09-27 12:09:13.313 UTC. Source: Retained task session record. Type: Clarification reply.

Original-message SHA256: d8ae0a676e5be0bf1f692cd41d53883c39d6eddf16bba9e5fb8f88754052147d

Question: What retention/deletion arrangement has been agreed for your P01 numeric gesture data, and have any of the outstanding second-laptop pause/restart, offline, or guide-only setup checks now been completed?

User answer: pending

PR-031

Recorded time: 2026-09-27 18:12:46.571 UTC. Source: Retained task session record. Type: User instruction.

Original-message SHA256: 0da9caac78076bfca9143f96afc2e390864fc3fac916bff7f0c6a12416453354

Go through the whole repo and do an audit sweep and delete any redundant code or files that are not related to my submission